

Cisco Team 3
CCNA-13: Integrated Enterprise Network Capstone
Sponsor Review Document

Trello:

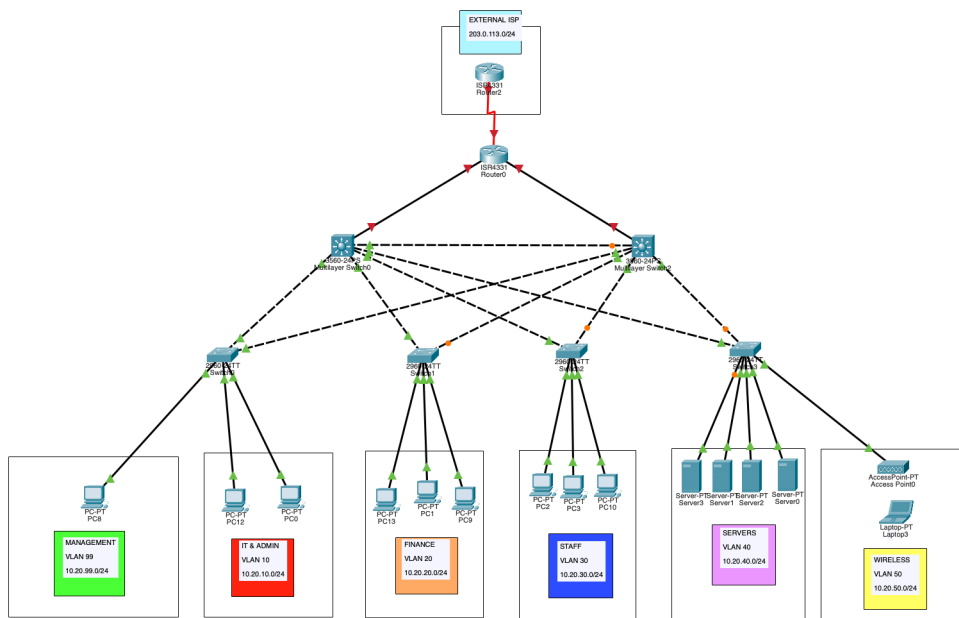
<https://trello.com/invite/b/6abd4fa5848c7a14efff92be/ATTIe677aebcd4b0511ddc7a58215ac4e6015EB5A6B2/it491-cisco-team-3-ccna-13>

GitHub: <https://github.com/benpopso/IT491-Team-3-Cisco-Team>

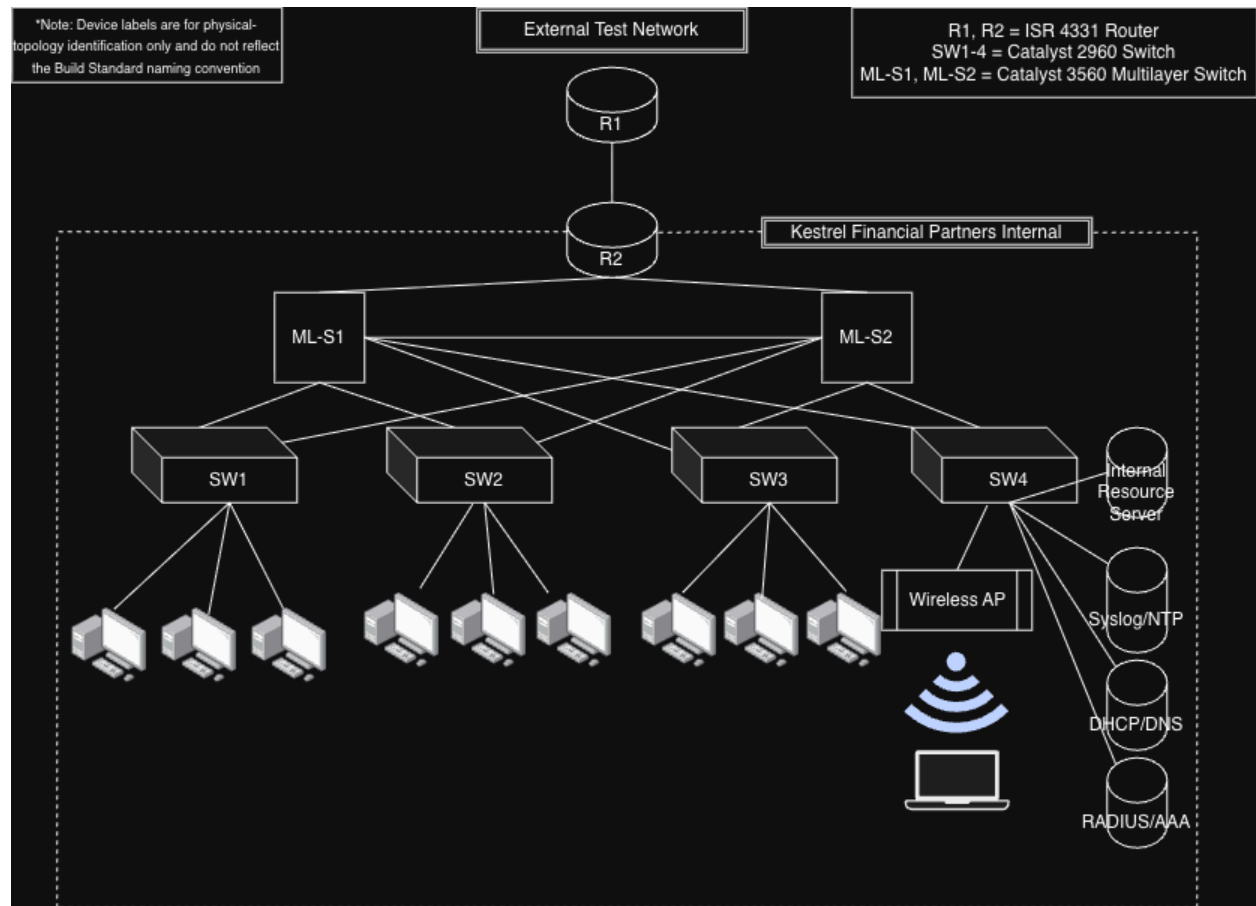
Purpose

This document presents Cisco Team 3's initial network design for sponsor review before implementation. It includes the proposed Packet Tracer layout, logical and physical topology, addressing plan, and initial network build standard. The design may be refined during implementation and capability testing as needed. We have several separate documents and diagrams that show this information, but we assembled a single document for easier viewing. Because of this, diagrams may appear a bit small, so you might need to zoom in, or Ben (PM) can send the articles individually.

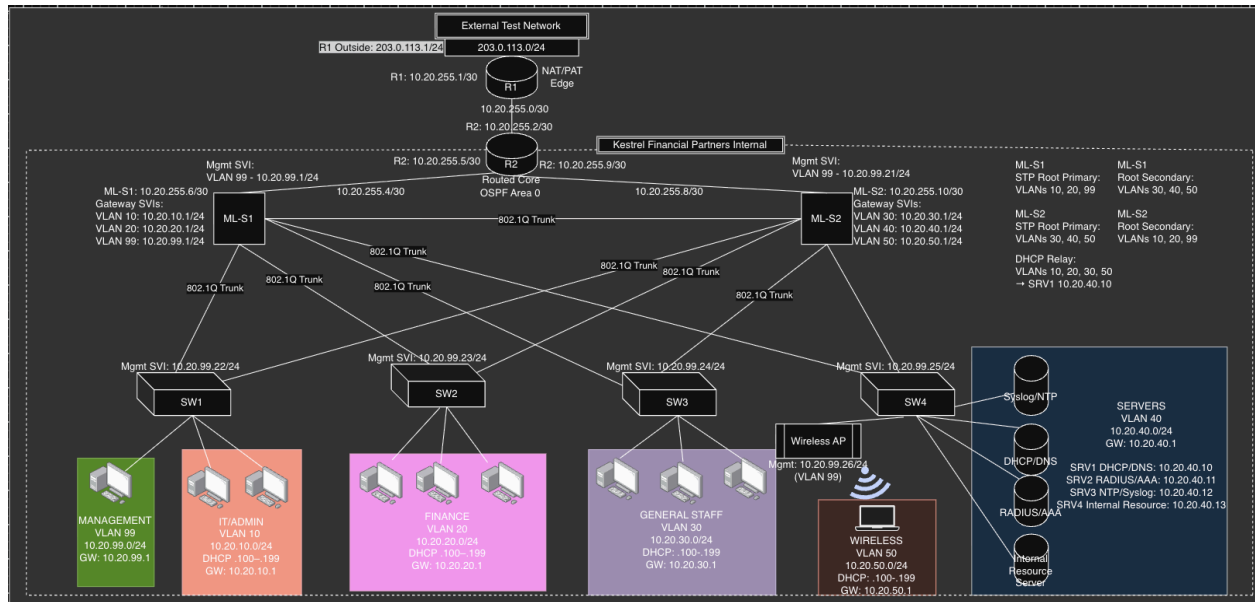
Initial Sketch in Packet Tracer:



Physical Design Diagram:



Logical Design Diagram:



Addressing Plan:

Cisco Team 3		Addressing Plan						
VLAN / Segment	Purpose	Network	Prefix	Subnet Mask	Default Gateway	Addressing / Interface Assignment	Broadcast	Gateway Device / SVI
VLAN 10	IT / Admin	10.20.10.0	/24	255.255.255.0	10.20.10.1	DHCP: 10.20.10.100-199	10.20.10.255	ML-S1
VLAN 20	Finance	10.20.20.0	/24	255.255.255.0	10.20.20.1	DHCP: 10.20.20.100-199	10.20.20.255	ML-S1
VLAN 30	General Staff	10.20.30.0	/24	255.255.255.0	10.20.30.1	DHCP: 10.20.30.100-199	10.20.30.255	ML-S2
VLAN 40	Servers	10.20.40.0	/24	255.255.255.0	10.20.40.1	Static	10.20.40.255	ML-S2
VLAN 50	Wireless	10.20.50.0	/24	255.255.255.0	10.20.50.1	DHCP: 10.20.50.100-199	10.20.50.255	ML-S2
VLAN 99	Management	10.20.99.0	/24	255.255.255.0	10.20.99.1	Static	10.20.99.255	ML-S1
R1 ↔ R2	Point-to-Point Transit	10.20.255.0	/30	255.255.255.252	N/A	R1: 10.20.255.1 R2: 10.20.255.2	10.20.255.3	N/A
R2 ↔ ML-S1	Point-to-Point Transit	10.20.255.4	/30	255.255.255.252	N/A	R2: 10.20.255.5 ML-S1: 10.20.255.6	10.20.255.7	N/A
R2 ↔ ML-S2	Point-to-Point Transit	10.20.255.8	/30	255.255.255.252	N/A	R2: 10.20.255.9 ML-S2: 10.20.255.10	10.20.255.11	N/A
External Test Network	Simulated External Network	203.0.113.0	/24	255.255.255.0	N/A	R1 outside: 203.0.113.1 (proposed)	203.0.113.255	N/A
Address Allocation								
Host Range	Intended Use	Reason						
.1	Default gateway / SVI	Consistent gateway convention						
.2-9	Reserved infrastructure	Leaves room for future infrastructure						
.10-19	Servers / network services	Predictable static service addresses						
.20-49	Network infrastructure	Switches, APs, other managed devices; gateway-owning devices may use their gateway SVI						
.50-99	Reserved static addresses	Future static devices						
.100-199	DHCP client pool	Dynamic employee/client endpoints						
.200-254	Reserved / future expansion	Prevents exhausting the planned structure						
Static Server Addressing								
Device	Service	VLAN	IP Address	Prefix	Default Gateway			
SRV1	DHCP / DNS	40	10.20.40.10	/24	10.20.40.1			
SRV2	RADIUS / AAA	40	10.20.40.11	/24	10.20.40.1			
SRV3	NTP / Syslog	40	10.20.40.12	/24	10.20.40.1			
SRV4	Internal Resource Server	40	10.20.40.13	/24	10.20.40.1			
Management Addressing								
Device	Role	Management Interface	Management IP	Note: This is set to .1 because it also has the SVI default gateway for VLAN 99				
ML-S1	Multilayer Switch	VLAN 99 SVI	10.20.99.1					
ML-S2	Multilayer Switch	VLAN 99 SVI	10.20.99.21					
SW1	Access Switch	VLAN 99 SVI	10.20.99.22					
SW2	Access Switch	VLAN 99 SVI	10.20.99.23					
SW3	Access Switch	VLAN 99 SVI	10.20.99.24					
SW4	Access Switch	VLAN 99 SVI	10.20.99.25					
AP1	Wireless Access Point	VLAN 99 Mngmt Inter	10.20.99.26					
DHCP Plan								
VLAN	Clients	DHCP Pool	Default Gateway	DNS Server				
10	IT/Admin	10.20.10.100-199	10.20.10.1	10.20.40.10				
20	Finance	10.20.20.100-199	10.20.20.1	10.20.40.10				
30	General Staff	10.20.30.100-199	10.20.30.1	10.20.40.10				
50	Wireless	10.20.50.100-199	10.20.50.1	10.20.40.10				
Addressing Design Note: The project-provided VLAN networks keep their assigned /24 scope boundary. Static infrastructure, network services, and DHCP client ranges are separated using a consistent host-addressing convention. The project-provided 10.20.255.0/24 routed transit allocation is subdivided into /30 point-to-point networks for the router links between R1, R2, ML-S1, and ML-S2. This provides a unique Layer 3 subnet and two usable interface addresses for each point-to-point connection. R1 serves as the NAT/PAT edge router toward the simulated external network, while R2 supports routed-core connectivity between the multilayer distribution environment and the edge.								

Initial Baseline:

General Overview

This document establishes the network build standard for the Integrated Enterprise Network Capstone. The purpose of it is to provide a consistent baseline for device naming, management access, security, centralized services, and general network configuration. All devices should follow the same approved standard for the network. This way, it remains organized, secure, and easier for the team to test and troubleshoot.

This standard represents the team's initial configuration baseline. It may be modified as implementation, integration testing, and Packet Tracer capability testing identify necessary changes. Approved revisions become the standard for subsequent implementation, testing, and auditing.

Naming

Device Type	<u>Naming Standard</u>
Edge Router	KFP-R1
Core Router	KFP-R2
Multilayer Switches	KFP-MLS1, KFP-MLS2
Access Switches	KFP-SW1 through KFP-SW4
DHCP/DNS Server	KFP-DHCP-DNS
RADIUS/AAA Server	KFP-RADIUS
Syslog/NTP Server	KFP-SYSLOG-NTP
Internal Resource Server	KFP-RESOURCE
Wireless Access Point	KFP-AP1

Interface and Cabling Standard

Connection Type	Interface / Cabling Standard
Router ↔ Router	Routed Ethernet Connection
Router↔Multilayer Switch	Routed Ethernet connection; multilayer-switch interface configured as a Layer 3 routed port
Multilayer Switch↔Multilayer Switch	802.1Q Ethernet trunk
Multilayer Switch↔Access Switch	802.1Q Ethernet Trunk
Access Switch↔PC/Server	Ethernet access port assigned to the appropriate VLAN
Access Switch↔Wireless AP	Ethernet access connection configured for AP/network plan
Wireless AP↔Wireless Client	802.11 wireless connection

Device-Type Baseline

Device Type	Requirements
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R1 Edge Router	Approved hostname; routed interfaces; single-area OSPF; NAT/PAT for outbound connectivity; SSH management; AAA/RADIUS with local fallback; configuration saved
R2 Core Router	Approved hostname; routed interfaces; single-area OSPF; SSH management; AAA/RADIUS with local fallback; configuration saved
KFP-MLS1/KFP-MLS2	Approved hostname; ip routing; assigned gateway SVIs according to Logical Topology; VLAN 99 management SVI; routed connection toward R2; 802.1Q trunks; single-area OSPF; assigned STP root primary/secondary roles; SSH/AAA management; configuration saved
KFP-SW1 through KFP-SW4	Approved hostname; VLAN 99 management SVI; 802.1Q uplink trunks; access ports assigned to appropriate VLANs; PortFast and BPDU Guard on applicable endpoint ports; port security on applicable endpoint ports; unused ports shutdown; SSH/AAA management; configuration saved
DHCP/DNS Server	Static server addressing per Addressing Plan; DHCP and DNS services enabled; DHCP scopes configured for applicable client VLANs; internal DNS records configured as required
RADIUS Server	Static server addressing per Addressing Plan; RADIUS/AAA service enabled; network devices configured as RADIUS clients where required; administrative and wireless authentication supported according to project requirements
Syslog/NTP Server	Static server addressing per Addressing Plan; NTP and Syslog services enabled; infrastructure devices use centralized time/logging where supported
Internal Resource Server	Static server addressing per Addressing Plan; internal service enabled as required for connectivity testing; used as protected permit/deny target
Wireless AP	Approved hostname/name where supported; management addressing per Addressing Plan; Wireless VLAN connectivity; secure wireless authentication; RADIUS-backed authentication where supported
PCs/Clients	Connected to assigned VLAN; DHCP used where

	specified by Addressing Plan; correct gateway/DNS received; used for permitted/denied connectivity testing
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Management Access

Network infrastructure will use the dedicated Management VLAN for in-band device management. Exact management addresses and interface assignments will follow the Addressing Plan.

- SSH will be used for remote device administration
- Unencrypted remote protocols such as Telnet will not be permitted
- RADIUS/AAA will provide centralized administrative authentication where supported
- Management access will be restricted according to the approved segmentation and ACL policy
- Devices of the same type will use consistent management-access settings

Switching and Routing Standard

The network uses 6 defined VLANs for IT/Admin, Finance, Wireless, General Staff, Servers, and Management. VLAN IDs, subnet assignments, gateway assignments, management addresses, and routed transit addressing will follow the Addressing Plan.

- Access-switch uplinks to multilayer switches will use 802.1Q trunking
- Connections between multilayer switches will use 802.1Q trunking
- Inter-VLAN routing will be performed by the multilayer switches using SVIs
- Gateway SVI ownership will follow the Logical Topology diagram
- Routed infrastructure links will use Layer 3 routed interfaces
- Internal dynamic routing will use single-area OSPF Area 0
- STP root bridge assignments will follow the Logical Topology and align with gateway ownership
- Required VLANs will be configured consistently across applicable switches
- Only required VLANs will be permitted across trunk links where practical

Access-Layer Hardening Standard

- Port security will be configured on applicable endpoint access ports
- PortFast will be enabled on applicable endpoints access ports
- BPDU Guard will be enabled on applicable PortFast/access ports
- Unused access ports will be administratively disabled
- Trunk ports will be explicitly configured as trunks where required

Services Standard

Centralized services will follow the server roles and static addressing defined in the Addressing Plan

- DHCP/DNS: DHCP will provide correct address range, default gateway, and DNS information for applicable client VLANs. DHCP relay will be configured where clients are separated from the centralized DHCP server by a Layer 3 boundary. DNS will provide internal name resolution
- RADIUS/AAA: The centralized RADIUS service will support authentication as required by the project. Device administration will use AAA
- NTP/Syslog: NTP will be established before Syslog is relied upon so centralized logs use consistent and trackable timestamps

- NAT/PAT: R1 will provide NAT/PAT for outbound connectivity to the simulated external network
- Wireless: Wireless clients will use the Wireless VLAN and receive addressing through DHCP. Wireless traffic will rely on RADIUS authentication as per project requirements

Device Security Standard

- Privileged EXEC access will be protected using enable secret
- A local admin account will be maintained as a fallback if RADIUS server is unavailable
- Console and VTY access will require authentication
- Admin sessions will use an inactivity timeout to limit indefinitely open sessions
- An authorized access warning banner will be configured on devices

ACL and Segmentation Standard

- Extended ACLs will enforce inter-segment access according to the segmentation plan
- Required traffic will be explicitly permitted where appropriate
- Restricted traffic will be denied according to the segmentation policy
- Access relationships will be tested in both permitted and denied directions
- ACLs must not unintentionally block required DHCP, DNS, or RADIUS communication
- ACL configuration will remain consistent with the approved VLAN and addressing design
- Exact ACL entries and placement may be refined during implementation and testing